

BIOLOGY 101 · ANSWER KEY - PROFESSOR COPY

Homework 1: Cells

Grading guide. Total **26 points**. Answers in different words that mean the same thing get full credit; partial credit is suggested where it applies. Enter the final score in Professor mode on the recitation page.

Part A — Label the cell

1. For each number on the diagram, write the part's name and its job (in your own words — the 'city' comparison is fine). (8 pts)

Answer: 1 Membrane — the wall/gatekeeper; decides what comes in and out. 2 Cytoplasm — the jelly that fills the cell. 3 Nucleus — the library; holds the DNA instructions. 4 Mitochondria — power plants; burn sugar with oxygen for energy. 1 point per name, 1 per job.

Part B — The ladder

2. Put these in order from SMALLEST to BIGGEST: organ, atom, cell, molecule, tissue, organism. (3 pts)

Answer: atom, molecule, cell, tissue, organ, organism. 3 points all correct; 2 points for one swap; 1 point for two swaps.

Part C — Doubling

3. One cell divides, and every new cell keeps dividing. Fill in how many cells there are after each round. (3 pts)

Answer: 1, 2, 4, 8, 16, 32, 64. 3 points all correct; 2 points for one slip; 1 for two.

4. After 10 rounds there are 1,024 cells. About how many after 11 rounds? (1 pt)

Answer: 2,048 (double 1,024).

Part D — Match the cell to its fact

5. Draw a line from each cell type to the fact that describes it. (4 pts)

Answer: Red blood cell -> Throws out its nucleus to make room for hemoglobin; lives about 120 days | Skin cell -> Stacks in layers to build a wall — only a few layers thick in a preemie | Bacterium -> A single-celled living thing; some kinds cause infections like sepsis | Nerve cell -> Carries electrical signals around the body (1 pt each)

Part E — Power plants

6. When mitochondria burn sugar with oxygen, what is the “exhaust”? (1 pt)

Answer: A. Carbon dioxide and water - Sugar + oxygen -> energy, with CO₂ and H₂O as exhaust — exactly the two things you breathe out more of than you breathe in.

Part F — Flashback to NICU Tutor

7. Why do newborns turn yellow (jaundice)? Use the words “red blood cells” and “bilirubin” in your answer. (2 pts)

Answer: Worn-out red blood cells get recycled, leaving a yellow leftover called bilirubin. Newborns recycle lots of extra red cells while their liver is still slow at flushing bilirubin out, so it builds up and tints the skin. 1 point each for using the two ideas correctly.

8. The isolette keeps the air very humid for the tiniest babies. What does that protect against? (1 pt)

Answer: Water (and heat) escaping through the baby's very thin skin — evaporation.

Part G — Think like a doctor

9. A preemie in the NICU is gaining about 20 grams every day. Using what you know about cells, explain why this baby needs so much sugar and oxygen. (3 pts)

Answer: Growing means cells dividing (doubling) — making new cells needs energy, and the cell's mitochondria make energy by burning sugar with oxygen. More growth = more dividing = more fuel needed, gram for gram, than an adult. 3 points for connecting growth -> cell division -> mitochondria need sugar + oxygen; 2 for two of the three links; 1 for one.